A Decodable Signalling Code Co-evolves but Stays Private to Kin: Survival Selection, Clonal Descent, and Why No Public Code Bootstrapped in a Minimal World

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Abstract

We ask whether a decodable signalling code can *grow* from random weights under survival selection alone (no gradient on the task, no oracle labels, no borrowed weights), and whether what grows is a public shared convention or a private kin code. In a minimal hand-coded 2-D lethal-vent world, tiny tied code-table agents evolve a decodable communication channel, and we decompose the result. Both arms carry the full code-table; the only manipulated variable is the reproduction-fitness signal (survival vs. uniform-random, with the reproduction RNG byte-matched between arms). The tied-architecture-plus-clonal-descent baseline alone (the random-fitness control, which also fixates to ~one lineage) reaches mutual intelligibility (MII) 0.1174, already ~73× the $1/625 \approx 0.0016$ chance floor; at the pre-registered gen-28 fixation point survival selection adds a paired win-margin of **+0.0548** on top (95% CI [0.0234, 0.0849], n=48; survival arm MII 0.17219 vs. random-fitness control 0.1174). Selection therefore *sharpens and tunes* a largely descent-determined channel rather than originating it from nothing. A secondary, post-hoc transient probe (n=24, checkpoints at gen 28/56/100/150) shows this margin is a **transient acceleration**, not a sustained equilibrium difference: the survival-arm MII stays ~stable (~0.16–0.17) while the random-fitness control's MII rises ($0.117 \to 0.138$) as clonal fixation alone accumulates cross-agent decodability, so the paired margin decays from +0.0569 [0.0162, 0.1021] (gen 28) through +0.0489 [0.0151, 0.0821] (gen 56) to +0.0329 [−0.0014, 0.0687] (gen 100) and +0.0249 [−0.0072, 0.0571] (gen 150). Within-kin decodability is thus fundamentally a clonal-descent phenomenon that survival selection merely *accelerates*. The code stays private to *kin*: the survival arm fixates to a single founder lineage (mean dominant-lineage share 99.1%, N_{eff} 1.02), so within-founder intelligibility runs ~0.16 (~100× chance) while cross-founder intelligibility sits at the chance floor. Three heterogeneous interventions that triangulate cross-lineage pressure all leave cross-founder intelligibility at chance; under these regimes a public cross-lineage code did not bootstrap (positive-or-inconclusive tier, not banked as a formal negative). The co-evolution headline is scoped to this world, this architecture, and a 28-generation timescale. Methodologically, we contribute a bidirectional self-audit whose single claim is **bidirectionality**: it is built to catch deflation as well as inflation. In this project it caught a pilot/formal false *negative* and an RNG-offset bug in the headline control, whose fix left the conclusion intact and slightly strengthened the margin ($+0.045 \to +0.0548$).

Code & data: all source, the hand-coded world, the experiment runners, and every banked verdict JSON are openly available at `https://github.com/yangchunxuan/grown-grounded-language`; every result regenerates via the per-stage commands in the Appendix and `offscreen/RUNBOOK.md`. An archival snapshot is deposited at Zenodo, DOI

https://doi.org/10.5281/zenodo.21074235 (concept DOI, always resolving to the latest version).

Keywords: artificial life; emergent communication; signalling code; survival selection; clonal descent; kin selection; adversarial self-audit methodology

1 Introduction

Where does a shared signalling code come from? A communication channel is *decodable* when a receiver can recover a signal's referents from the signal alone, and a code is *public* when distinct lineages (not just kin) can decode one another. Most computational accounts of how such a code arises take one of two routes. In the signalling-game and reinforcement tradition, communication is *trained in*: a channel is optimised by gradient descent on a task with a differentiable channel (Foerster et al., 2016), or by reward on a referential task that supplies a task signal each round (Lazaridou et al., 2017). In the iterated-learning tradition, structure is *induced by design*: a transmission bottleneck is imposed between generations so that only learnable, compressible codes survive (Kirby et al., 2008). Both routes answer a real question, but neither asks the more basic one we take up here: can a decodable signalling code **grow** (neither designed-in nor trained-in) from random initial weights under **survival selection alone**, with no gradient on the communicative task, no oracle or truth labels, and no borrowed or pretrained weights? And if such a code does grow, is it a **public** code shared across the population, or a **private** code confined to kin?

We study this question in a deliberately minimal substrate: a hand-coded, fixed-rule 2-D lethal-vent world in which agents metabolise, can die, and reproduce, and in which a tiny tied code-table (the only communicative apparatus) is initialised from $\mathcal{N}(0,1)$ noise. The world is not learned and the channel is not optimised; the sole pressure on communication is that organisms that use it must, on average, survive and reproduce more. Mutual intelligibility (MII) is measured not by weight similarity but as inter-agent codebook agreement: a sampled set of referents is passed through one agent's emission and another's decoding, and we score exact-tuple recovery against a chance baseline of $1/625 \approx 0.0016$. This minimality is the point: a small world makes the mechanism legible and lets us build a control that is matched to the experimental arm in every respect except the reproduction-fitness signal.

The scientific contribution is a single, scoped finding with three parts. First, within this world and over a 28-generation timescale, a decodable channel **co-evolves from scratch**, but the result decomposes into a descent baseline and a selection increment. The control (the **random-fitness control**) also fixates to ~one lineage, so its MII of **0.1174** (already ~73× the 1/625 chance floor) is the tied-architecture-plus-clonal-descent baseline, not zero. At the pre-registered gen-28 fixation point, survival selection adds **+0.0548** on top (95% CI [0.0234, 0.0849], n=48). A secondary post-hoc transient probe shows this gen-28 margin is a **transient acceleration** of clonally-shared decodability rather than a sustained equilibrium difference. Second, this code is **not public**. The survival arm fixates to a single founder lineage (mean dominant-lineage share **99.1%**, N_{eff} **1.02**), so the co-evolved code is a **kin code**. Third, actively maintaining lineage diversity is *necessary but not sufficient*. In the one sustained-diversity cell (soft96, n=8) coexisting non-kin lineages still decode one another only at the floor, and putting cross-lineage decoding directly under selection does not bootstrap a public code: higher pressure only crashes survival. We name this a **pressure-vs-survival tension**.

The second contribution is methodological: a reusable self-audit whose single methodological claim is **bidirectionality**, that it catches *deflation* as readily as inflation. We demonstrate this with two in-scope catches reproducible from this paper's own artifacts: a pilot/formal false *negative* and an RNG-offset bug in the headline random-fitness control (whose fix slightly strengthened the margin, +0.045 → +0.0548).

2 Related Work

Our study sits at the intersection of five lines of work and is shaped by a sixth: a methodological critique about over-claimed emergence and weak baselines.

Emergent communication in MARL. A large body of deep multi-agent reinforcement-learning work shows that communication protocols arise when agents are jointly optimized on a task, typically with a differentiable or otherwise privileged channel (Foerster et al., 2016), or are scored by reward on a referential task that supplies a task signal on every round (Lazaridou et al., 2017). Grounded compositional language can emerge in multi-agent populations under such task-optimized training (Mordatch and Abbeel, 2018), and the modern deep-emergent-communication literature is broad enough to have its own reviews and tooling (Lazaridou and Baroni, 2020; Kharitonov et al., 2019). Our setup deliberately removes these affordances. There is no gradient on the channel and no oracle: adaptation proceeds by mutation and selection on survival alone.

Signaling and convention games. The game-theoretic tradition, from Lewis's signaling games (Lewis, 1969) to Skyrms's evolutionary treatment (Skyrms, 2010), establishes that stable signaling conventions can emerge from local dynamics without a designer, and that multiple incompatible conventions can be locally stable. Our notion of a private kin code is consistent with the existence of mutually incompatible conventions. We offer a mechanistic instance of the convention-multiplicity that signaling theory predicts, and measure inter-agent codebook agreement directly.

Kin selection and signalling among relatives. Our headline outcome is a *kin*-private code, which places it squarely in the kin-selection tradition: relatedness lowers the threshold for cooperative and communicative traits to be favoured, and signalling among relatives can be stable where signalling among strangers is not (Hamilton, 1964). In our world a private code is selectively cheap precisely because the survival population collapses to ~one founder lineage.

Iterated learning and compositionality. A prominent account of how structured, transmissible language arises is iterated learning, in which a *transmission bottleneck* pressures languages toward compositional, generalizable structure (Kirby et al., 2008). Neural iterated learning realizes this pressure in deep agents and can yield compositional structure (Ren et al., 2020). At the same time, emergent-language compositionality is not automatically tied to generalization, and agent languages can be effective without being human-like or compositional (Kottur et al., 2017; Chaabouni et al., 2020). Our design takes a pointed stance: we do **not** impose a transmission bottleneck. Rather than building in the mechanism known to produce shareable structure, we ask whether one is *needed*.

Artificial-life and embodied evolved communication. Within ALife, the evolution of communication has been studied through simulated agents under selection (Cangelosi and Parisi, 2002) and through situated, grounded language-game and robotic models (Steels, 2003; Quinn, 2001; Marocco and Nolfi, 2007). Our work is squarely in this lineage in spirit but is distinguished by an explicit, architecture-matched control.

The replication-hygiene critique. A recurring concern is that "language emerged" claims are often confounded by gradient or oracle leakage, lack a matched control, and rest on underpowered single-seed runs; relatedly, emergent protocols are often degenerate or fail without explicit pressure (Kottur et al., 2017; Chaabouni et al., 2020). This critique motivates our second contribution: a reusable adversarial self-audit combining an architecture-matched control, multi-seed bootstrap-CI win-margins, frozen pre-registration, and a no-oracle redline.

Positioning. The field reliably produces public, grounded, or compositional communication, but in lanes that supply exactly what our setting withholds: gradient-MARL with task rewards and interacting partners (Mordatch and Abbeel, 2018; Lazaridou and Baroni, 2020), a deep-learning infrastructure for emergent-language games (Kharitonov et al., 2019), imposed iterated-learning bottlenecks (Ren et al., 2020), or populations of large language models that already possess a borrowed, pre-trained language (Flint Ashery et al., 2025). The LLM line is a contrast case only, not a mechanism we build on. Our question is therefore not whether public communication can be engineered (it can) but the origins/sufficiency question those lines do not address: whether a decodable, grounded signalling code can grow from random weights under survival selection alone, with no gradient on communication, no oracle, and no borrowed weights.

3 World and Agents

The substrate is deliberately minimal and entirely hand-coded. Nothing in the world or the agent body is learned; the only adaptive object is each agent's communication code-table, and even that adapts by mutation and selection rather than by any gradient on the survival task.

3.1 The RTC lethal-vent world

The world (`rtc_world.py`, with constants frozen in `config/prereg_rtc.py`, `RTC_VERSION = "RTC-MVP0-rich-world-v1"`) is a 48×48 grid carrying a 4-channel scalar field. The field evolves by a fixed-rule physics: advection, diffusion, per-step decay, and injection from a small number of point vents per channel. These rules are simulator code, not a trained model: the world dynamics never change in response to agent behavior.

An agent does not read the field directly. Perception passes through a no-oracle seam: agents receive only lagged, windowed, noisy local sensor readings plus their own body state and any decoded messages, never the ground-truth patch value, edibility, or any founder identity used as a policy input. The input to each agent is a *lagged, noisy* sensor read, which is why we keep our claims at "**decodable signalling code**" rather than "grounded": the channel can be decoded and is behaviourally useful, but we do not assert that its symbols track world state per se.

Survival is metabolic. Each agent carries an energy reserve and pays a fixed upkeep every step. When it eats, the edible value of a patch is a dot product between the local 4-channel patch values and a fixed per-agent metabolic weight vector; the same patch is nutritious to one metabolism and useless or toxic to another. A value below the toxic floor is immediately lethal. Death is therefore a hard, hand-coded consequence of foraging in a flowing field—the selection pressure the rest of the study rides on.

3.2 The agent and its code-table

The agent body is intentionally trivial so that communication is the only thing under evolutionary pressure. The communicating object is a single `UnifiedIO` tied code-table of shape $K \times \text{BANDS} \times \text{VOCAB} = 4 \times 5 \times 6$, initialized from $\mathcal{N}(0, 1)$. The table is *tied*: the same parameters drive both speaking and listening, so a mutation that changes how an agent emits also changes how it decodes. A speaker emits by taking, per channel, the argmax over the VOCAB symbols; a listener predicts by taking, per channel, the argmax over the BANDS value-bins. This coupling is load-bearing for the convergence argument: emit and predict cannot drift apart independently (see Figure 6).

We quantify communication with mutual intelligibility (MII): a set of referents is probed through one agent's `emit` into another agent's `decode`, scoring exact recovery of the full tuple. The referent space is the band×channel product, so chance recovery is $1/5^4 = 1/625 \approx 0.0016$. Within-founder (WF) and cross-founder (CF) MII are read off the off-diagonal of the full population×population MII matrix, filtered by lineage.

3.3 Reproduction, and the from-scratch invariants

Reproduction is mutation plus selection, with no gradient on the survival task. Offspring are produced by adding Gaussian noise to the parent's tied code-table. Selection has two pre-registered variants used across the arc: hard top-50% truncation, and a soft tournament (k=2) on lineage-shared fitness for niching.

The from-scratch invariants are strict: (i) there is *no gradient on the survival task*; (ii) the *only* gradient mechanism anywhere in the codebase is an optional cultural-imitation helper, and it is *unused in every load-bearing arm* reported here; (iii) there are *no borrowed* *or pretrained weights and no language model.*

4 Methods: The Attack-Machine

Every claim in this paper is the output not of a single experiment but of a procedure built to surface our own errors before review does. We call this procedure the **self-audit**: six interlocking disciplines. Its single methodological claim is **bidirectionality**: a self-audit should catch *deflation* as readily as inflation.

4.1 The mutual-intelligibility (MII) metric

The unit of evidence is **inter-agent codebook agreement**: can one agent decode what another emits? For an ordered pair of agents, we sample a fixed set of referents, route each through the speaker's `emit` and the listener's `predict`, and score **exact-tuple recovery**. With the code-table tied at $K \times \text{BANDS} \times \text{VOCAB} = 4 \times 5 \times 6$ and a referent set of 625, chance is $1/625 \approx 0.0016$. This unforgiving, no-partial-credit operationalization means an above-chance MII reflects a reconstructed *symbol-to-referent mapping*, not two networks that happen to look alike.

We compute MII at two scopes. **Within-founder MII (WF)** pairs agents from the same founder lineage; **cross-founder MII (CF)** pairs agents from distinct founder lineages coexisting at the same generation. The WF/CF split is what lets us distinguish a *private kin code* (WF high, CF at chance) from a *public code* (CF also above chance).

To guard against MII being a measurement artifact, every stage runs two pre-logged sentinels. A **lineage-shuffle** control permutes lineage labels: if the metric truly keys on lineage, CF should spike under shuffling—it does (CF $0.0009 \to 0.065$ in the diagnostic; 15–23$\times$ across C1 cells). A **gate-0 content test** compares open messaging against muted and scrambled channels: open $\gg$ mute/scramble with CIs excluding zero, establishing that message *content* is load-bearing.

4.2 The architecture-matched random-fitness control

Our control is **not** communication-ablated: **both arms carry the full `UnifiedIO` code-table**, the same initialization, and the identical evolutionary loop. The **only** manipulated variable is the **reproduction-fitness signal** (survival in the treatment arm versus uniform-random in the control), with the reproduction RNG **byte-matched between arms**. We therefore call it the **random-fitness control**.

Crucially, this control isolates *selection for communication* from **clonal descent-convergence**: the random-fitness arm **also** fixates to ~one lineage, so its MII of **0.1174** is the tied-architecture-plus-clonal-descent baseline (~73× the floor), not zero. At gen 28 the survival arm reaches MII **0.17219**, a **paired** win-margin of **+0.0548, 95% CI [0.0234, 0.0849]**, at n=48.

This control also carries the project's two hardest lessons, both caught by the self-audit (§6). First, a *formal* treatment was once compared against a *pilot* control, manufacturing a false **negative**. Second, a multi-model adversarial review found that an earlier version drew its fake fitness from the *shared* reproduction RNG; fixing it and re-running at n=48 left the conclusion intact and slightly *strengthened* the margin (+0.045 → +0.0548).

4.3 Multi-seed bootstrap-CI win-margins

We report no single-seed positives. Every load-bearing comparison runs $\geq$ 6 seeds (the headline uses 48), and the quantity reported is a **win-margin with a bootstrap confidence interval over per-seed paired differences**. A claim counts as supported only when the paired margin's CI excludes zero in the predicted direction. The headline check is the random-fitness control: the survival arm beats it by the paired win-margin **+0.0548, 95% CI [0.0234, 0.0849]** at n=48 (§5.1).

Multiple comparisons. Each stage has a single, pre-registered **primary endpoint** locked before the run. The sentinel checks (lineage-shuffle, gate-0 content, founder-quality control) are **secondary corroborating diagnostics**, not independent tests of the headline.

We apply **no family-wise correction** by design: there is a single locked primary per stage. The "pressure sweep" of §5.4 is a *post-hoc synthesis* of three separately pre-registered probes; we treat it as triangulation, not as a corrected family.

4.4 Frozen pre-registration

Each stage's specification—hypotheses, coexistence gates, outcome labels, and sample size—is **locked in a versioned spec before the run**. Gates and labels are not moved after seeing data. The discipline's value is most visible when it costs us something: the diagnostic's frozen gate was met by only 5/16 seeds, forcing the pre-registered verdict `INCONCLUSIVE_NO_WINDOW` even though the *direction* (private codes) was unanimous and strong.

4.5 The no-oracle redline

The central failure mode in this literature is leakage: an agent that secretly reads world truth can fake communication. We draw a hard **no-oracle redline**: agent decisions may read **only decoded messages, own body state, and legal sensors**, *never* true world patch, edibility, target-code, or founder-id-as-policy-input. A static-analysis scan (`_redline_scan`) enforces a subset on the core co-evolution runner `rtc_g1f_coevolve.py` only; the cross-lineage runners (C2X/C2X2/C2X3) are enforced by construction and checked by code-grounded review, not by this automated scan.

4.6 Code-grounded adversarial multi-agent review

Before any spec is locked, it is attacked by independent reviewers reading the *actual runner code*, not a prose summary. This caught, e.g., the C2X "content-free" auto-label as a viability-crash mislabel, and forced two reviewer contradictions in C1 to be adjudicated on the record before locking.

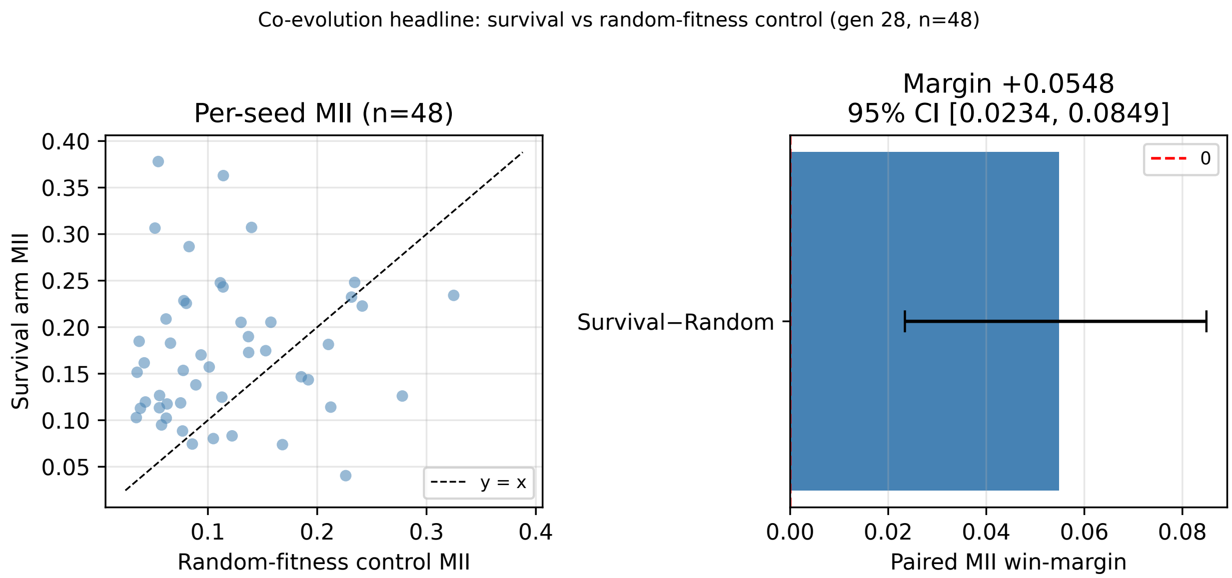

Figure 1: Co-evolution headline (n=48, gen 28). **Left:** per-seed MII scatter: survival arm vs. random-fitness control; points above the diagonal indicate a per-seed win. **Right:** paired win-margin +0.0548 (95% CI [0.0234, 0.0849]), CI excludes zero. Source: `rtc_g1f_commblind_verdict_formal48_rngfix.json`.

4.7 The tier rule and the provenance triple

The **tier rule** (frozen): a run at $n \leq 8$ may only conclude POSITIVE or INCONCLUSIVE, never a scoped negative; a negative is banked only at $n = 16$. Accordingly, the cross-lineage CF-at-floor result is reported as "a public code did not bootstrap under the regimes tested" *with a stated mechanism, across seeds and across three heterogeneous interventions*, but explicitly **not as a formal negative**.

Every result self-describes via a **provenance triple**: the immutable verdict JSON at a git commit, the CLAIM_LEDGER mapping each claim to its verdict-file and commit, and the RUNBOOK giving the exact regeneration command per stage.

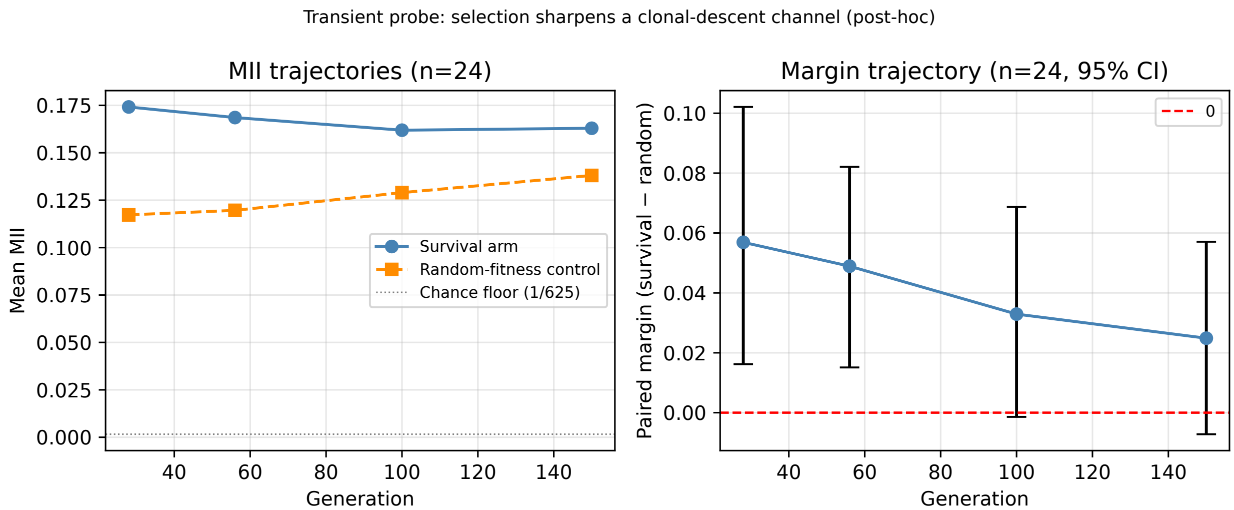

Figure 2: Transient probe (n=24, post-hoc). **Left:** MII trajectories of both arms across generations 28/56/100/150. The random-fitness control rises as clonal fixation accumulates. **Right:** paired margin with 95% CI decays monotonically; CI excludes zero at gens 28 and 56, includes zero at gens 100 and 150. Source: `rtc_g1f_transient_probe_verdict.json`.

5 Results

5.1 Co-evolution is real, but kin-scoped

The first question is the existence question: does a decodable communication channel co-evolve from random weights under survival selection alone? In the g1f configuration of the lethal-vent world, the answer is yes, but the scope of that "yes" is the load-bearing part of the result.

The headline (a decomposition). Under formal settings, the co-evolved channel reaches a mean cross-agent MII of **0.17219**, against the random-fitness control at **0.1174**. This isolates *selection for communication* from **clonal descent-convergence**: the random-fitness arm also fixates to ~one founder lineage, so its MII is not zero but the descent/architecture baseline that any heritable code-table reaches. That baseline is already **0.1174, ~73× the 1/625 chance floor**. At the pre-registered gen-28 fixation point, survival selection then adds a **paired** bootstrap win-margin of **+0.0548, 95% CI** [**0.0234, 0.0849**] (n=48; see Figure 1). The reading is therefore a decomposition rather than an origination: the tied architecture plus clonal descent reaches MII 0.1174 on its own, and survival selection **sharpens and tunes** that channel by +0.0548.

Robustness: the margin is a transient acceleration (secondary, post-hoc). The +0.0548 is the **pre-registered gen-28 primary** endpoint and it stands at that timescale. A secondary, post-hoc transient probe (n=24, checkpoints at gen 28/56/100/150) shows the paired margin **decays monotonically**: gen 28 margin +0.0569, CI [0.0162, 0.1021] (excludes 0); gen 56 +0.0489, CI [0.0151, 0.0821] (excludes 0); gen 100 +0.0329, CI [−0.0014, 0.0687] (includes 0); gen 150 +0.0249, CI [−0.0072, 0.0571] (includes 0). The mechanism is visible in the two arms separately (see Figure 2): the survival-arm MII stays roughly stable (~0.16–0.17) while the **random-fitness control's MII rises** (0.117 → 0.138) as single-lineage clonal fixation alone keeps accumulating cross-agent decodability. The reading is therefore that survival selection produces a **transient acceleration** of clonally-shared decodability rather than a sustained equilibrium difference, and this is the empirical justification for scoping the headline to 28 generations.

The scope (load-bearing). The survival arm fixates to a single founder lineage: across n=48 seeds the mean dominant-lineage share is **99.1%** (95% CI [97.3%, 100%]), the inverse-Simpson effective number of lineages $\mathbf{N}_{\text{eff}} = 1.02$, and **47 of 48** seeds end essentially single-lineage. The co-evolved channel is therefore **kin-lineage decodability, a private kin code, not a public cross-lineage shared convention**.

5.2 Why kin-only? (diagnostic)

To distinguish whether the kin-only signature is an artifact of fast founder collapse (hypothesis A: collapse-limited) or genuine private-code divergence (hypothesis C: private codes), we pre-registered a comparison under a frozen gate.

The frozen verdict is `INCONCLUSIVE_NO_WINDOW`: only **5/16** seeds opened a coexistence window. We report this as inconclusive rather than relabeling the gate after seeing the data.

The **direction**, however, is strong and unanimous, pointing to **C** (see Figure 3). Across

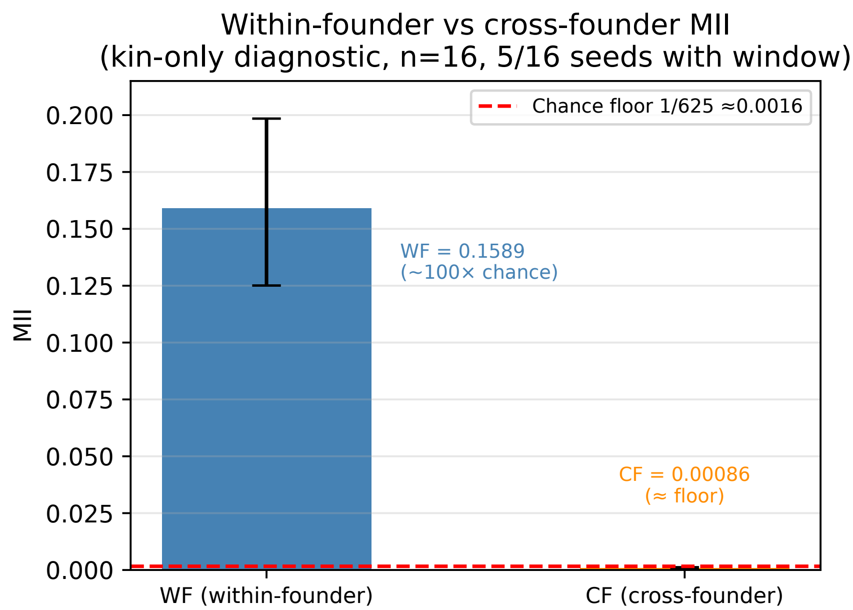

Figure 3: Within-founder (WF) vs. cross-founder (CF) MII with chance floor (kin-only diagnostic, n=16, 5/16 seeds with coexistence window). WF ≈ 0.159 (≈ 100× chance); CF ≈ 0.00086 (≈ floor). Source: `rtc_g1f_kinonly_diagnostic_verdict.json`.

the measurable seeds, within-founder MII is **0.159** (≈ 100× chance) while cross-founder MII is **0.00086** (≈ chance); the ratio CF/WF is **0.5%**, and all **5/5** measurable seeds take the same form: whenever two non-kin lineages coexisted, they decoded each other essentially at chance while each decoded its own kin ≈ 100× above chance.

Two pre-registered sentinels rule out the obvious artifacts. (i) **Lineage-shuffle**: permuting lineage labels lifts CF from 0.0009 to 0.065 (a ~75× jump), confirming the metric truly keys on lineage membership. (ii) **Gate-0 (content load-bearing)**: open-channel populations survive far better than muted or scrambled ones (fraction alive 0.31 vs. 0.03 vs. 0.05, CIs excluding zero), so message content is load-bearing for survival. The floor-level CF is therefore genuine cross-lineage unintelligibility, not an inert channel.

5.3 Maintaining diversity is necessary but not sufficient

We ran a 2 × 2 factorial—selection {hard top-50% truncation; soft = tournament-k2 with lineage fitness-sharing} × population {16, 96}—to test whether actively sustaining diversity

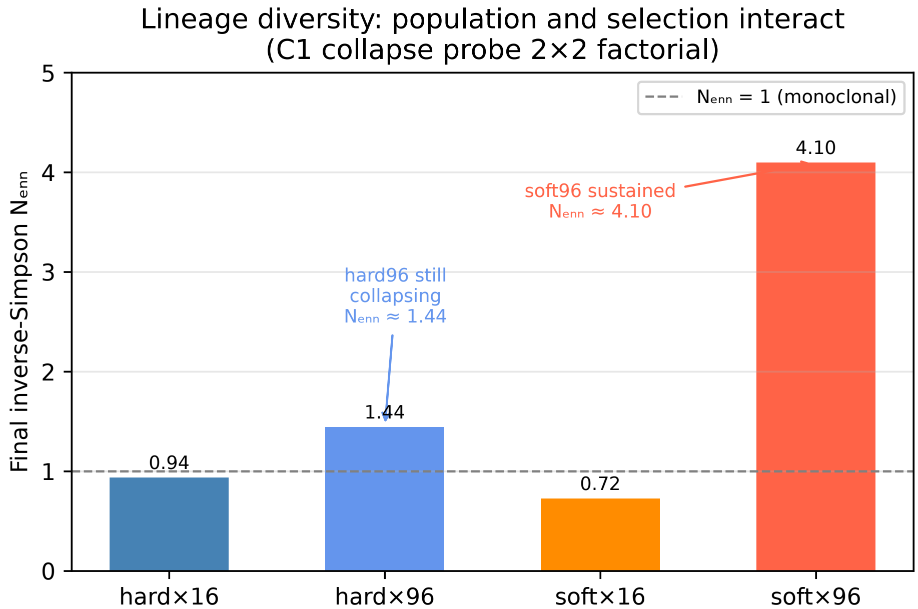

Figure 4: Final inverse-Simpson N_{eff} across the 2×2 factorial (C1 collapse probe). Population size opens windows; only soft (niching) selection sustains diversity. Soft×96: N_{eff} ≈ 4.10 (sustained); hard×96: N_{eff} ≈ 1.44 (still collapsing). Source: `rtc_g1f_c1_collapse_probe_verdict.json`.

lifts CF off the floor.

Table 1: C1 collapse probe: 2×2 results. WF ≈ 0.16 across viable cells; CF at floor (≈ 0.0016) throughout. All CF−FLOOR CIs straddle or exclude zero from below.

Cell	N_{eff}	CF	WF	CF/WF
hard×16	0.94 (collapsed)	0.00081	0.174	0.5%
hard×96	1.44 (collapses)	0.00176	0.175	1.0%
soft×16	0.72 (collapsed)	0.00072	0.096	0.8%
soft×96	**4.10 (sustained)**	0.00153	0.160	1.0%

Mechanism: population opens windows, niching sustains them (see Figure 4). At pop=16 the population collapses regardless of selection (final N_{eff} 0.94 hard, 0.72 soft). At pop=96 windows open in all 8/8 seeds, but only soft selection *sustains* the diversity: hard96 still collapses to a final N_{eff} of 1.44, whereas soft96 ends at N_{eff} 4.10 in all 8 seeds, holding 15–25 coexisting generations each.

Consistent with C, against the strong form of A (directional, n=8). Within soft96 (the one cell that actually delivers sustained multi-lineage coexistence), cross-founder

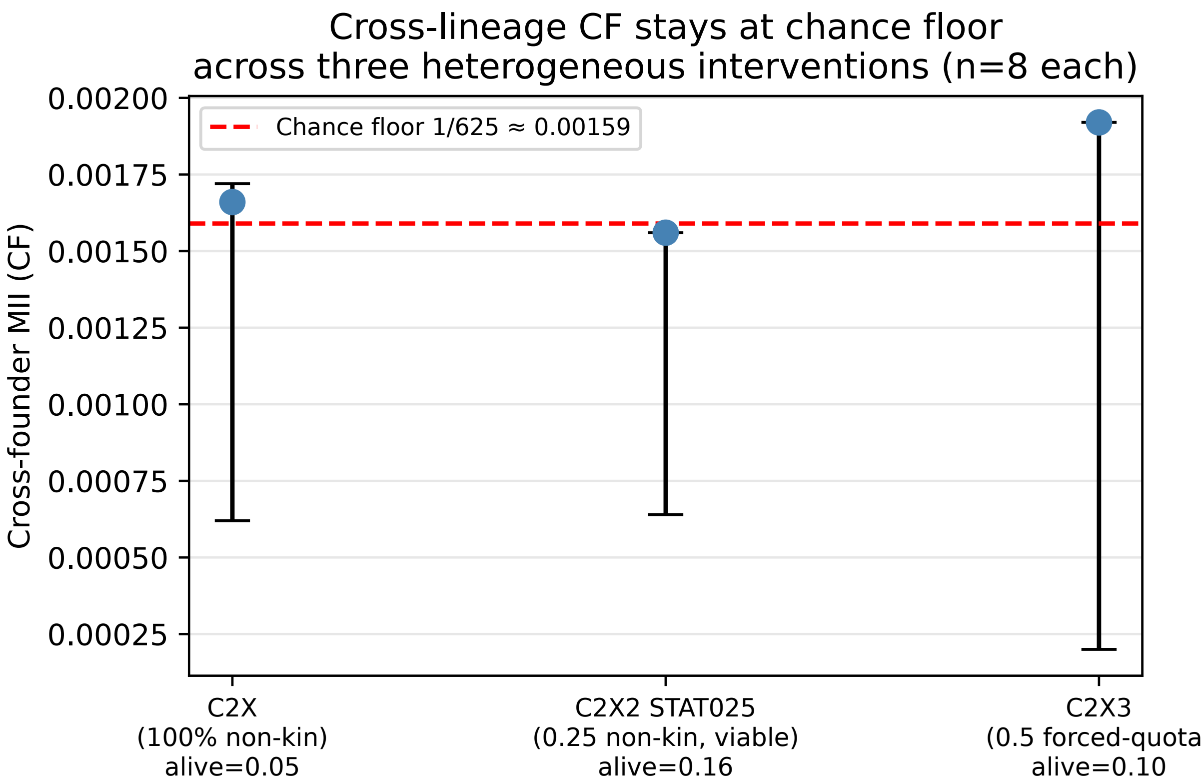

Figure 5: Cross-lineage CF stays at or below the chance floor across three heterogeneous interventions (n=8 each). The viable cell (C2X2 STAT025, alive $\approx$ 0.16) is the load-bearing case; C2X crashes survival (alive 0.045); C2X3 boom-bust (alive 0.10). Sources: `rtc_g1f_c2x*_verdict.json`.

MII still sits at the floor: CF 0.00153 versus floor 0.0016, with the CF−FLOOR confidence interval [−0.00031, +0.00021] straddling zero, and CF/WF $\approx$ 1% against a within-founder MII of $\approx$ 0.16 ($\approx$ 100$\times$ chance). Preventing collapse and sustaining 20+ generations of genuine non-kin coexistence did **not** lift CF off the floor. Maintaining diversity is **necessary** but, on this evidence, **not sufficient** for a shared, public code.

5.4 The wall: no public code bootstraps

Section 5.1 established a decodable channel that co-evolves but stays kin-private. We then attacked the obvious rescue: if survival depends on decoding non-kin, would a public code be selected up? We ran **three heterogeneous interventions** (C2X, C2X2, C2X3; different designs and scaffolds, 40 vs. 48 generations), each locked only after code-grounded adversarial review. They are **not** a single graded dose-response sweep; rather they **triangulate the same null** from three directions.

The three interventions. The **viable cell (C2X2 stationary scaffold, 0.25 non-kin)** is the load-bearing one because the population stays alive: mean fraction alive 0.1574, clearing the pre-registered viability bar, with coexistence windows in all 8/8 seeds. Yet cross-founder MII (CF) sits at the floor: CF = 0.00156 against a chance/floor of 0.00159, a margin of -0.00065, 95% CI $[-0.00095, -0.00038]$ (CI excludes zero), while within-founder MII (WF) is 0.156. The **forced-quota intervention (C2X3, 0.5 non-kin)** corroborates: mean alive 0.0998, below the viability bar, and CF never leaves the floor (seed-aggregate CF = 0.00192, margin -0.00086, CI $[-0.00139, -0.00032]$). The **cold 100%-non-kin intervention (C2X)** shows the viability failure mode: mean alive collapses to 0.0455 versus the kin-only reference of 0.2718 and the population dies before selection can act on communication (CF_OPEN = 0.00166, margin CI $[-0.00097, 0.00013]$ includes zero).

Headline. Across all three interventions, *cross-founder MII never rose above chance* (see Figure 5). Turning the cross-lineage screw did not buy a public code; where it bit at all, it crashed survival.

Mechanism. The **pressure-vs-survival tension**: non-kin codes are mutually undecodable, so an agent forced to listen to non-kin signals receives, in effect, noise, and acting on noise costs foraging time. A viable, diverse, *converging* regime is never reached for long enough to matter.

Power tier. All arms here are n=8—the POSITIVE-or-INCONCLUSIVE tier. We therefore do not bank a FORMAL negative. We report that **a public code did not bootstrap under the regimes tested**: CF stays at the architecture floor across all seeds *and* across all three heterogeneous interventions, with a stated mechanism. This triangulation across heterogeneous probes is qualitatively stronger than an isolated underpowered null. But per our calibration discipline, we claim only that under these regimes a public code *did* *not* bootstrap, never that this architecture *cannot* produce one.

6 The Methodology, Demonstrated

The preceding sections used the self-audit to defend a set of positive and null claims. This section turns the lens around. The audit's distinctive value, its single methodological claim, is that it runs **bidirectionally**: it is built to catch its operators inflating a result, and it is equally built to catch them *deflating* one. We anchor this on **two in-scope, reproducible catches**: a pilot/formal false *negative*, and the RNG-offset bug in the headline random-fitness control.

It caught a false negative (§5.1). The g1f line began with a committed "clean negative": survival selection did not appear to co-evolve a decodable channel. The self-audit exposed this as an artifact: a *formal* co-evolution arm (28 generations) had been compared against a *pilot* control. The load-bearing knob was generations (the effect plateaus around generation 12–14, and the pilot had truncated mid-rise at 7). Re-run with both arms formal and paired at the same initialization, the survival arm beats the random-fitness control by an MII win-margin of **+0.0548, 95% CI [0.0234, 0.0849]** at n=48. A real, peer-review-relevant effect had been suppressed; the audit recovered it.

It caught an RNG-offset bug in the headline control. A multi-model adversarial review found that the random-fitness control drew its fake fitness from the *shared* reproduction RNG, partially coupling the two arms. Fixing it (an independent generator for the fake fitness) and re-running at n=48 left the conclusion intact and **slightly strengthened** the margin, from +0.045 to **+0.0548** (survival arm 0.17219, control 0.1174, CI [0.0234, 0.0849]).

It held a frozen gate to an honest inconclusive (§5.2). The kin-only diagnostic pre-registered a coexistence gate. Only 5 of 16 seeds cleared it (founder collapse was too fast), so the pre-registered verdict returned `INCONCLUSIVE_NO_WINDOW`. The frozen gate held; we report inconclusive on the window and direction-only on the lean.

It adjudicated two direct contradictions between independent reviewers (§5.3). Locking the C1 collapse-probe spec required resolving two disagreements between independent code-grounded reviewers (the correct fitness-sharing divisor and the coexistence-window gate definition). The spec was locked (v5.1) only after the contradictions were explicitly adjudicated. The machine is the forum in which reviewer disagreement is forced to a decision before any number is generated.

It corrected an over-coarse auto-label using a pre-logged continuous metric (§5.4). The C1 window classifier emitted the binary label `POP_DRIFT`, which undersells the mechanism. The pre-logged continuous N_{eff} trajectory refined it: population size *opens* windows, but only soft/niching selection *sustains* diversity. Logging a continuous metric before knowing the answer let us correct a label that a coarse gate alone would have shipped.

7 Discussion

7.1 "Decodable but private to kin" and what it means for language-origin theory

The central finding of this arc is a dissociation that language-origin theory does not usually hold apart: a communication channel can be *decodable* without being *public*. Under survival selection alone, a decodable channel co-evolves from random weights and beats an architecture-matched random-fitness control (paired win-margin +0.0548, 95% CI [0.0234, 0.0849], n=48), with the caveat that the control already reaches 0.1174 (~73× chance) from clonal descent alone, so selection *sharpens* rather than originates the channel, and this gen-28 margin is a transient acceleration the control's own clonal fixation closes by ~gen 100. Within-founder mutual intelligibility (WF) sits near 0.16, roughly 100× the $1/625 \approx 0.0016$ chance floor, and the content is load-bearing. Across every regime tested, cross-founder MII (CF) never rose above chance. The code is real and it means something—but only to kin.

For theories of language origin, this matters because intelligibility is often treated as the hard part and publicness as a near-automatic consequence. Our minimal world separates the two. A decodable code emerged readily; publicness did not emerge at all. The reason is mechanistic: non-kin codes are mutually undecodable, and within a single lineage there is no selective pressure to be understood by strangers one will, in practice, almost never meet (the formal population fixates to ~99% a single founder lineage, N_{eff} 1.02). A private kin code is, in this world, an *attractor*.

What is non-obvious, relative to kin-selection and iterated-learning theory. Hamilton (1964) makes a *kin-private* outcome unsurprising once relatedness is high; iterated-learning theory predicts that *shareable* structure needs a transmission bottleneck (Kirby et al., 2008). Two things in our result are not handed to us by either theory. First, a *decodable* channel emerged **without being trained or designed in**. Second, the privacy we report is shown as "founders never converge": independently seeded lineages starting at ≈zero cross-founder intelligibility and staying there through 20+ actively-sustained coexisting generations (C1 soft96). That a *sustained-diversity* regime still yields no convergence is the non-trivial claim.

7.2 Why a public code plausibly needs more world, not more selection

The natural rescue is to *select for* cross-lineage understanding: make survival depend on decoding non-kin. We did exactly this across three heterogeneous interventions. It did not work, and the way it failed is informative. In the viable 0.25 cell the population stayed alive (alive ≈ 0.16) but cross-founder MII held at floor; under the 0.5 forced-quota intervention survival became boom-bust (mean alive 0.10) with CF never exceeding 0.0058 across all seeds and generations; under cold 100% non-kin the population simply crashed (alive 0.045 vs. 0.272). Where the pressure bit at all, it only crashed survival harder. We call this the **pressure-vs-survival tension**: forcing agents to listen to non-kin whose codes they cannot yet decode degrades foraging and kills them before selection can act on communication.

This points away from "more selection pressure" and toward "more world." Two ingredients appear to be missing: first, **shared referents worth naming**; second, **sustainable diversity under communication pressure**. A public code presupposes multiple lineages that both persist *and* benefit from understanding one another. We showed that maintaining diversity is necessary but not sufficient: niching (tournament selection plus lineage fitness-sharing) sustained genuine multi-lineage coexistence at pop=96 (soft96 final N_{eff} 4.10, 8/8 seeds, 15–25 coexisting generations), yet cross-founder MII still sat at the floor.

7.3 Relation to the iterated-learning bottleneck

We deliberately did *not* impose a transmission bottleneck (Kirby et al., 2008). Our agents inherit weights by mutation and selection, not by learning from a teacher through a narrow channel. The pressure-vs-survival tension we observe can be read as suggesting *why* such a bottleneck, or an ecological analogue of one, may be a prerequisite for a public code: without a mechanism that repeatedly forces codes through a shared, learnable interface, lineages drift into mutually private dialects and there is no force pulling them back together. We frame this as a hypothesis the arc *motivates*, not a result it establishes.

7.4 Honest limits

These claims are scoped narrowly and on purpose. They hold for **one hand-coded world** (the RTC lethal-vent substrate), **one architecture** (the tied UnifiedIO code-table, random $\mathcal{N}(0,1)$ init), and **per-stage timescales** (28 generations for the headline co-evolution, kin-only diagnostic, and C1 arms; 40–48 generations for the cross-lineage pressure probes). The headline co-evolution result is well powered (n=48), though its absolute margin is modest and decomposes into a clonal-descent baseline (MII 0.1174) plus a selection increment (+0.0548) that a secondary, post-hoc transient probe (n=24) shows is a **transient acceleration** that decays to a CI including zero by ~gen 100. The cross-lineage outcome is *not* a formal negative: a public code "did not bootstrap under the regimes tested (n=8, positive-or-inconclusive tier), across three heterogeneous interventions, with a stated mechanism." We make no impossibility claim.

Why 28 generations suffices for the bootstrap question. A reviewer may reasonably ask whether running pop=96 for thousands of generations would let mutation eventually break the kin-private wall. We argue it would not. First, bootstrapping a public code is an *initiation* question, not an *equilibrium* one: a public code that was going to emerge would first show cross-founder MII lifting off the floor during sustained coexistence and then amplifying. In the soft96 cell we sustained 20+ generations of genuine multi-lineage coexistence (N_{eff} 4.10) and CF never lifted even partially, so there is no nascent signal for additional generations to amplify. Second, no mechanism supplies a force toward cross-lineage alignment: independently seeded code-tables occupy a 625-referent space in which mutual intelligibility by drift alone is a near-measure-zero coincidence (chance $\approx 1/625$ per referent), and selection actively *opposes* alignment through the pressure-vs-survival tension. Additional generations add drift, not a new directional force. Third, the regime is computationally bounded: our formal cross-lineage runs already cost ~30–60 min each at pop=96 over 40–48 generations, so the 10^3–10^4-generation, multi-seed runs a confidence interval would require are prohibitive, and, by the first two points, would probe a regime in which theory predicts no change. We therefore bound the claim to the timescales run and frame the long-run outcome as expected-null but empirically untested: a public code *did not* bootstrap under the regimes tested, not that none *could* over astronomically longer runs.

8 Conclusion and Future Work

This paper makes one coupled claim with two faces. **Scientifically**, in a minimal hand-coded 2-D survival world, with no gradient on the task, no oracle or truth labels, and no borrowed or pretrained weights, a *decodable* communication channel **co-evolves from random weights under survival selection alone**, but the result decomposes. The architecture-matched

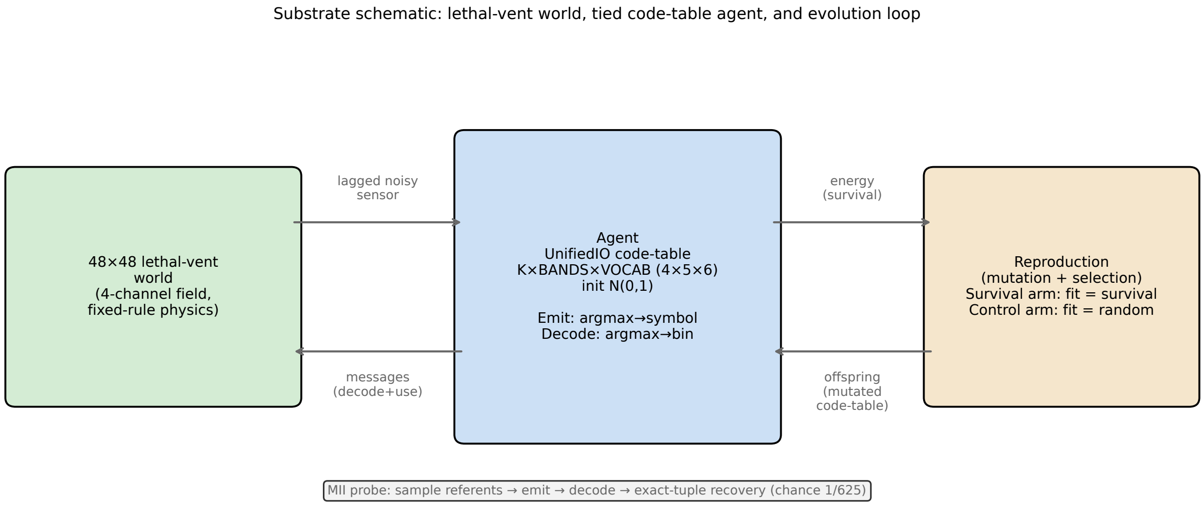

Figure 6: Substrate schematic: 48×48 lethal-vent world, the tied `UnifiedIO` code-table agent ($4 \times 5 \times 6$, $\mathcal{N}(0, 1)$ init), and the mutation-plus-selection reproduction loop. MII is measured by sampling referents, routing through `emit`→`decode`, and scoring exact-tuple recovery (chance 1/625).

random-fitness control already reaches MII **0.1174** (~73× the 1/625 chance floor) from clonal descent alone; at the pre-registered gen-28 fixation point survival selection then **sharpens** this channel by a paired MII win-margin of **+0.0548 (95% CI [0.0234, 0.0849],** n=48**)** (see Figure 6), a margin a secondary post-hoc probe (n=24) shows is a **transient acceleration** that decays to a CI including zero by ~gen 100. The co-evolved code remains a **private kin code**: the survival arm fixates to a single founder lineage (dominant-lineage share 99.1%, N_{eff} 1.02), within-founder mutual intelligibility reaches ≈ 0.16 ($\approx 100\times$ chance) while cross-founder intelligibility sits at chance. When diversity is actively sustained (niching at pop=96, soft96, N_{eff} 4.10 over 20+ generations), cross-founder intelligibility still does not leave the floor. Across **three heterogeneous interventions** (C2X, C2X2, C2X3) that triangulate cross-lineage pressure, cross-founder intelligibility *never rose above chance*; where the pressure bit, it only crashed survival harder. We scope every part of this to *this world and this architecture*, and to each stage's timescale. Under these regimes a public code *did not* bootstrap; we explicitly *decline to bank this as a formal negative* (n=8 positive-or-inconclusive tier).

Methodologically, this arc is more believable because of the bidirectional self-audit that produced it. Its single claim is **bidirectionality** (catching deflation as well as inflation), anchored on two in-scope, reproducible catches: a pilot/formal false *negative* (§5.1) and an RNG-offset bug in the headline random-fitness control surfaced by a multi-model adversarial review (§6), whose fix slightly *strengthened* the margin (+0.045 → +0.0548).

Future work. The arc is self-contained; it *motivates rather than requires* its sequel. The central next question is sharp: **is the public-code bottleneck the world/ecology, or the optimizer?** We name two candidate interventions, both required to be oracle-clean and not gradient-on-survival: (a) a richer world with spatial demes and a resource ecology that naturally sustains coexisting lineages and provides a real survival payoff for decoding non-kin; (b) consequence-driven lifetime learning in which agents learn from their *own* survival outcome, never supervised on the true referent. Neural iterated learning is an established from-scratch route to structure under an imposed transmission bottleneck (Ren et al., 2020); the open problem here is doing that under survival selection in a grounded world, without borrowing a pre-trained language or leaking the true referent.

A Reproducibility

Every result is anchored to three durable, reachable handles: (i) the **verdict JSON**, identified by filename and present both in the repository and in the archived `v2.0-g1f` release (Zenodo DOI `https://doi.org/10.5281/zenodo.21074235`); (ii) the claim ledger (`offscreen/CLAIM_LEDGER.md`), mapping each claim to its verdict file; and (iii) the runbook (`offscreen/RUNBOOK.md`), giving the exact command, env vars, and effective config per stage. The public git history was subsequently normalized to a single author, so historical commit hashes are not reachable in the current repository; verify each result by its verdict filename in the repository, the `v2.0-g1f` release, or the Zenodo deposit, and regenerate via the runbook. (Each verdict JSON's embedded `provenance.git_commit` likewise refers to the pre-normalization history.)

Run commands per stage

All commands are issued from the repository root. `RTC_G1F_FORMAL=1` selects the formal configuration; `RTC_G1F_FORMAL=0` is a fast pilot used only for smoke testing. The lethal-world constant `RTC_TOXIC_DEATH=-0.9` is set automatically by every g1f harness via `os.environ.setdefault`; all banked g1f-arc runs use `-0.9`.

- **g1f (survival-selected channel; 28 generations):** reproduce as a documented multi-step sequence. (1) Run the n=48 RNG-fixed random-fitness control: `RTC_G1F_FORMAL=1 RTC_TOXIC_DEATH=-0.9 RTC_G1F_COMMBLIND_SEEDS=48 python -m offscreen.rtc_g1f_commblind_control`, banked as `rtc_g1f_commblind_verdict_formal48_rngfix.json`. This rngfix verdict is the **sole source** for every headline number. (2) The legacy helpers (`.._formal48.json`, `.._reconciled_verdict.json`) predate the RNG-fix and are **superseded**. (3) Lineage composition (`RTC_G1F_LINEAGE_SEEDS=48 python -m offscreen.rtc_g1f_lineage_share`): `rtc_g1f_lineage_share_verdict.json`: dominant-share 99.1%, N_{eff} 1.02.

- **g1f transient probe (secondary, post-hoc; gen 28/56/100/150, n=24):** `rtc_g1f_transient_probe_verdict.json`.

- **Kin-only diagnostic (28 generations):** `rtc_g1f_kinonly_diagnostic_verdict.json`.

- **C1 collapse probe (2×2, 28 generations):** `rtc_g1f_c1_collapse_probe_verdict.json`.

- **C2X (forced 100% non-kin; 40 generations):** `rtc_g1f_c2x_crosslineage_verdict.json`.

- **C2X2 (ramped + stationary scaffold; 48 generations):** `rtc_g1f_c2x2_ramped_verdict.json`.
- **C2X3 (forced-diversity quota + 0.5 non-kin, 48 generations):** `rtc_g1f_c2x3_forced_verdict.json`.

CI invariants

Heavy runs cannot be CI-automated (GPU/CPU minutes-to-hours); they are version-controlled *recipes*, not pipeline steps. What *is* automated are the scale-independent harness invariants under `tests/`, enforced on every push by `.github/workflows/ci.yml` via `python -m pytest tests/ -q`. These guard against silent behavior drift: vectorized MII is byte-identical to the reference loop; `make_provenance` produces a well-formed record; all RTC modules import cleanly; and the no-oracle redline scan (`_redline_scan` over `rtc_g1f_coevolve.py`) returns zero hits. This automated scan covers the core co-evolution runner only; the cross-lineage runners (C2X/C2X2/C2X3) are enforced by construction and checked by code-grounded review, not by this scan.

Declarations

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